

■ PROJECT DETAILS (WITH AZURE SYNAPSE)

● Project Title

Data-Driven Food Waste Management System using Azure Data Engineering Stack

● Project Goal

Build an end-to-end data engineering pipeline using Azure services to analyze and reduce cafeteria food waste using structured, semi-structured, and unstructured data.

The system helps:

Reduce food waste

Improve preparation planning

Track attendance vs food demand

Monitor student feedback

Support data-driven decision making

● Business Problem

Cafeterias prepare food based on assumptions, leading to:

Excess food waste

Financial loss

Poor menu planning

Storage issues

This project uses data pipelines + analytics to solve this.

● Data Types Used

✓ Structured Data

Stored in Synapse Tables

Examples:

Attendance

Food Prepared Quantity

Food Consumed Quantity

Food Waste Quantity

Ratings

✓ Semi-Structured Data

Stored in ADLS (JSON)

Examples:

Menu Plan JSON

Feedback JSON

Kitchen Logs

Processed using Databricks.

✓ Unstructured Data

Stored in ADLS Blob

Examples:

Food waste images

Text feedback

Optional voice complaints

● Azure Services Used (Updated)

Service

Purpose

Azure Data Factory

Data ingestion

Azure Data Lake Storage

Data storage (Raw, Clean, Curated)

Azure Databricks

Data transformation

★ Azure Synapse Analytics

Data Warehouse + SQL Analytics

Power BI

Dashboard & Reporting

● End-to-End Data Pipeline (WITH SYNAPSE)

● Step 1 — Data Ingestion (ADF)

ADF loads:

CSV attendance data

JSON menu data

Feedback data

Image metadata

Into: 👉 ADLS Raw Layer

● Step 2 — Data Processing (Databricks)

Databricks performs:

Data cleaning

Unit conversion (grams → kg)

Join multiple datasets

KPI calculations

Output stored in: 👉 ADLS Curated Layer

● Step 3 — Data Warehousing (★ Azure Synapse)

Synapse is used for:

✓ Creating Data Warehouse Tables

✓ Star Schema Modeling

✓ SQL Analytics Queries

✓ Creating Views for Power BI

✓ Fast Reporting Queries

Example: Fact Tables → Waste, Consumption

Dimension Tables → Date, Food Item, Menu

● Step 4 — Visualization (Power BI)

Power BI connects to: 👉 Synapse SQL Views

Dashboards show:

Waste trends

Efficiency metrics

Menu performance

Attendance vs waste

● Why Azure Synapse Is Important (Interview Point 🔥)

You can say:

Synapse is used as the serving and analytics layer where curated data is stored in star schema format and exposed through SQL views for reporting and dashboarding.

● KPIs Calculated

Waste %

= Waste Qty / Prepared Qty × 100

Consumption Efficiency %

= Consumed / Prepared × 100

Waste Per Student

= Waste Kg / Attendance

● Testing Strategy

✓ Data Quality Tests

No null dates

No negative waste

Attendance > 0

✓ Pipeline Tests

ADF pipeline run success

Databricks job success

Synapse tables updated

✓ Reporting Tests

Power BI refresh success

KPI values valid

● Data Architecture Layers

Raw Layer — ADLS

Original data (CSV, JSON, Images)

Clean Layer — ADLS

Validated + standardized data

Curated Layer — ADLS + Synapse

Business-ready data + warehouse tables

● Expected Business Outcome

Reduced food waste

Better food planning

Cost savings

Improved student satisfaction

● Future Enhancements

ML-based demand prediction

Real-time streaming data

Image-based waste detection